

UNIT 2

Mathematical Reasoning

About the Test

The GED® *Mathematical Reasoning Test* evaluates your ability to solve mathematical problems and apply mathematics to a variety of contexts. You will have 90 minutes to answer approximately 46 questions based on the following:

- **Quantitative Reasoning (45%)**, which includes problems with positive and negative whole numbers, decimals, and fractions; ratios, proportions, and percents; data and statistics; and geometric measurement
- **Algebraic Reasoning (55%)**, which includes expressions, polynomials, equations, inequalities, linear equations, quadratic equations, and patterns and functions. A variety of formats will assess evaluating, solving, and graphing skills in algebra.

You will need a minimum score of 145 to pass the *Mathematical Reasoning Test*, which is one of the four tests you will need to pass in order to earn a high school equivalency diploma.

Tools

Calculator For most of the test, you will be able to use either a handheld or an online version of the **Texas Instruments TI-30XS MultiView™ scientific calculator**. You can access the online calculator by clicking on this icon. You can also take your own handheld TI-30XS MultiView™ calculator to use on test day.



For the first five questions on the test, you will not be able to use the calculator. You can take up to 12 of the 90 minutes with these questions. Check all of your work carefully before you leave this section of the test. You will not be able to return to these problems once the online calculator is enabled.

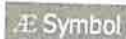
If you forget how to use a specific calculator function, you will be able to access a **Calculator Reference Sheet** from the computer screen by clicking on this icon.



Formula Sheet A copy of this formula sheet appears on page 413 of this book. You will be able to access these formulas by clicking on this icon. Other formulas that appear in this book should be memorized.



Symbol Selector On some fill-in-the-blank problems, you will need to use a symbol from the symbol selector toolbar. You can enable it when the icon to the right shows up on the light blue toolbar at the top of your screen.



At the testing center, you will be given a dry erase marker and a **wipe-off board** that you can use as scratch paper.

Computer-Based Mathematics Test Formats

Many of the questions will be *multiple-choice* with four options (A through D). You will **click** on the best answer.

A painter needs to cover the walls of four rooms in two coats of paint. Each room measures 4 ft by 8 ft, and the ceiling is 10 feet high in each. Each gallon of paint covers 32 square feet. How many gallons of paint does he need?

- ☐ A. 2
- ☐ B. 60
- ☐ C. 128
- ☐ D. 256

GED® TEST TIP

Follow all directions carefully. For example, in this fill-in-the-blank problem, you are asked to type the answer, simplify the expression, and use the symbol selector to complete the result.

Fill-in-the-blank questions will require you to **type** a numerical answer to a problem or to enter an equation by **clicking** to open the symbol selector and then **inserting** the symbol in the box. You may also be asked to **type** a word or phrase in the box to answer a question about your thought process.

Type your answer in the box. You may use numbers, symbols, and/or text in your answer.

Simplify the expression completely. Leave your answer in radical form.

$$\sqrt{60} \div \sqrt{5}$$

(NOTE: Click the symbol selector when you need to insert the radical sign.)

Drop-down questions give you the opportunity to **click** the word "Select" and choose the option that correctly completes the statement.

Sarah runs an open babysitting service on Saturday nights so that parents in the neighborhood can enjoy a night out. The chart below shows the number of children in attendance each hour.

Hour	# of children
5 p.m.	2
6 p.m.	4
7 p.m.	5
8 p.m.	6
9 p.m.	5
10 p.m.	2
11 p.m.	2

The number of children in attendance ranged from Select... to Select...

Drag-and-drop items require you to **click** on an image, number, or symbol and to **drag** and **drop** it to a specified location on the computer screen to create an expression, equation, or inequality.

A clown creates an equation that relates his data of each child's head size to the length of the balloon he needs to make them a proper balloon-animal hat. For each child, he needs a balloon five times as long as the child's head size, plus $\frac{4}{5}$ of an inch extra. Complete a linear equation for child A, where x is the size of the child's head in inches and y is the length of the balloon in inches.

Click on the variables and numbers you want to select and drag them into the boxes.

Equation for Child A

$$y = \square \square + \square$$

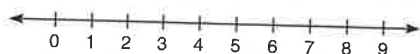
0.7	0.8	-0.7	-0.8
5	5.2	5.4	x

Hot-spot questions are based on a graphic such as a number line, coordinate grid, or geometric figure. You will need to **click** on a specific place or places on the figure.

Jessica has 20 hours a week to spend making scarves. Each week, she spends 2 hours purchasing the materials. Each scarf takes 3 hours to complete. The inequality $2 + 3s \leq 20$ represents the number of scarves, s , Jessica can make in one week.

Using the number line below, click each number to plot each possible value for s .

(NOTE: If you need to remove a point, hold the cursor over the point and left-click the mouse.)



GED® TEST TIP

You may need to click on one point or on more than one point to correctly answer a hot-spot question. Check the finished graphic against the question before moving on. Make sure that you have completed the task by creating all required points.

GED® TEST TIP

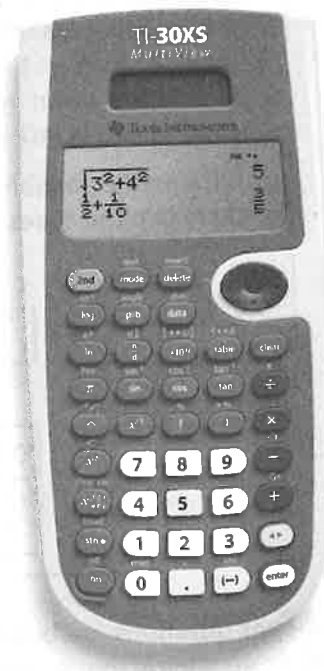
You are encouraged to purchase a handheld version of the Texas Instruments TI-30XS MultiView™ calculator for practice as you work through this book. You can buy one at office or school supply stores or online.

Using the Texas Instruments TI-30XS MultiView™ Calculator

Throughout this unit, you will practice using the TI-30XS MultiView™ calculator. Remember: You do not have to use the calculator to solve problems; only use it if it is relevant to the problem or if you believe it will be helpful.

Below is a picture of the handheld version and some issues to keep in mind.

- The handheld version has an On/Off button at the lower-left hand side.
- To use the online version of the calculator, click the calculator icon (see page 213) to open the calculator in a new window. Use your mouse to **drag** the calculator to the side of the computer screen so that it does not cover the problem. When you are finished with a calculation, click on the "X" at the top of the window to close the calculator.
- When you are finished with a calculation, use the **enter** button. (There is no "equal sign" button on this calculator.)
- Use the **white numeric keypad** to enter numbers, the decimal point, and the negative sign (–) for a negative number such as –13.
- The **operation keys**, on the right, allow you to add, subtract, multiply, and divide. (Use the minus sign here for subtraction; use the negative sign (–) on the keypad to indicate a negative number.)
- Use the bright green **2nd button** at the left of the calculator to activate the **2nd functions** (in bright green) above the keys. Some of the main ones are square root, exponent, percent, and mixed number functions.
- The **four arrows** at the top right of the calculator allow you to move the **onscreen cursor** up, down, left, or right as needed. The **delete** key to the left of the arrows allows you to correct mistakes as you work.
- Every time you begin a calculation, press the **clear** button above the **operation keys** to clear the calculator's memory.



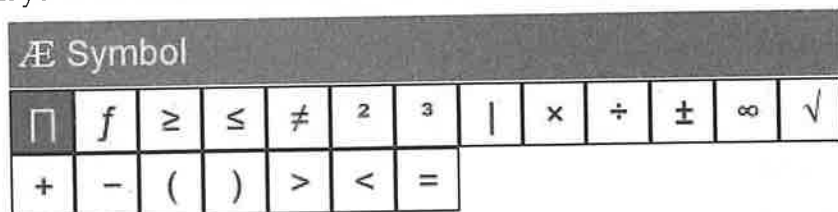
TI-30XS MultiView™ Calculator

Photo used with permission from TI. <http://education.ti.com>.

Accessing the Symbol Selector Toolbar

Some questions will instruct you to use a particular symbol from the “symbol selector.” This is not available for all problems, but this icon shows up when you need the toolbar: 

When you **click** on this icon, a new window will open up with these symbols:



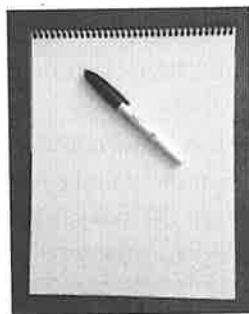
Click on a symbol to insert it in the correct place in an expression, equation, or inequality. When you are finished with the toolbar, **click** on the “X” at the top of the window to close the window.

Using the Wipe-Off Board

Even with all of the computer-based tools, you may still feel the need for some scratch paper to do the following:

- Write down the relevant numbers that you need to answer a specific question.
- Solve the problem by hand and use the calculator to check your work.
- Make a drawing or a diagram to help you picture a situation that is described in a word problem. This is an important strategy when working with measurement and geometry problems that do not contain a graphic.

At the testing center, you will be provided with a wipe-off board and dry erase marker, similar to the material shown at the right. Make use of the board throughout the *Mathematical Reasoning Test* and ask for a new one if you feel that you need it.



GED® TEST TIP

If you do not know all of these symbols now, copy them on a piece of paper or in a math notebook. Write meanings and examples for them as you encounter them in this book so you have mastery of them before the test.

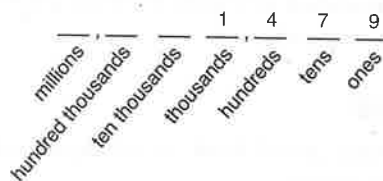
Number Sense and Problem Solving

Lesson 1: Compare and Order Numbers

Place Value

Numbers are part of your everyday life. Whether you're paying with cash, reading bus schedules, or changing television channels, you're using whole numbers.

The value of a number depends on the **place value** of its digits. On the place-value chart below, note that the value of a digit increases as you move to the left.



In the whole number shown on the chart, 1 has the greatest value. The number would be read as "one thousand four hundred seventy-nine."

Comparing and Ordering Values

To **order** numbers, you need to compare the value of their digits. Align the place values of the numbers you are comparing. Start at the left and compare the value of the first digit of each number.

Example 1: Place the numbers 342, 98, and 317 in order from least to greatest.

1. Compare the first digit in each number. Since 98 is a two-digit number, and the other numbers are three-digit numbers, 98 has the smallest value.
2. Next, compare the first digit of the remaining numbers. Since both numbers have the same digit (3) in the hundreds place, compare the digits in the tens place. Since 1 is less than 4, 317 is less than 342.

From least to greatest:
98, 317, 342

Rounding Numbers

Place value is also essential to **rounding** numbers.

Example 2: Round 2451 to the nearest hundred. 2451

1. Locate the place value that you want to round to. Then look at the place value to the right. If the digit to the right is 5 or greater, round up. If the digit to the right is less than 5, don't change the number in the hundreds place.
2. Round up the digit in the hundreds place. Then change the digits to the right of the hundreds place to zeros.

The value to the right of the hundreds place is a 5. Round up.

2451 rounds up to **2500**

Key Ideas

- When ordering numbers, compare the same place values between numbers.
- Round numbers to estimate answers or eliminate answer choices in multiple-choice questions.

GED® TEST TIP

It's important to know the inequality symbols below.

$<$ means "is less than"

$\leq$ means "is less than or equal to"

$>$ means "is greater than"

$\geq$ means "is greater than or equal to"

Practice 1

- A. Write the digit from the number below that corresponds to the listed place value. The first one is done for you.

1,436,879

- | | |
|-----------------------------------|-------------------------------|
| <u> 4 </u> 1. hundred thousands | <u> </u> 5. thousands |
| <u> </u> 2. hundreds | <u> </u> 6. tens |
| <u> </u> 3. ones | <u> </u> 7. ten thousands |
| <u> </u> 4. millions | |

- B. Round these numbers as directed.

- | | |
|--|---|
| 8. Round 544 to the nearest hundred. | 11. Round 11,632 to the nearest thousand. |
| 9. Round 76 to the nearest ten. | 12. Round 1525 to the nearest thousand. |
| 10. Round 1058 to the nearest hundred. | 13. Round 84 to the nearest hundred. |

- C. In each of the following pairs, which number is greater?

- | | |
|----------------|------------------------|
| 14. 100 or 89 | 16. 1099 or 1145 |
| 15. 339 or 341 | 17. 125,391 or 119,450 |

- D. Write these numbers in order from least to greatest.

- | | |
|-----------------------|--------------------------|
| 18. 23 18 45 39 | 21. 1429 1420 1432 1425 |
| 19. 111 89 109 91 | 22. 12,071 11,098 12,131 |
| 20. 1087 932 909 1139 | 23. 15,356 15,309 15,298 |

- E. Choose the one best answer to each question.

- | | |
|--|---|
| 24. When stacking items, the heaviest items should be placed at the bottom. Starting at the bottom, in what order should items weighing 45 pounds, 40 pounds, 50 pounds, and 48 pounds be stacked?
A. 40, 45, 48, 50
B. 45, 50, 40, 48
C. 50, 48, 45, 40
D. 50, 40, 45, 48 | 25. Which of the following correctly shows 1,543,976 rounded to the nearest hundred thousand?
A. 2,000,000
B. 1,600,000
C. 1,500,000
D. 1,540,000 |
|--|---|

Answers and explanations begin on page 661.

Lesson 2: Whole Number Operations

Key Ideas

- Add when combining amounts and subtract when finding the difference.
- Align place-value columns when adding or subtracting.
- Always look to the next place-value column to regroup.

GED® TEST TIP

Another way to check answers is to round the numbers in a problem and find an approximate answer.

Addition and Subtraction

You use **addition** when you need to combine amounts. The answer in an addition problem is called the **sum**, or **total**. When you are adding, it's helpful to stack the numbers in a column. Be sure to line up the place-value columns and then work from right to left, starting with the ones column.

Sometimes the digits in a place-value column add up to 10 or more. When this happens, you will need to **regroup** to the next place value.

Example 1: Add $40 + 129 + 24$.

1. Align the numbers you want to add on the ones column. Working from right to left, add the ones column first. Since the ones column totals 13, write the 3 in the ones column and regroup, or **carry**, the 1 ten to the tens column.

$$\begin{array}{r} 1 \\ 40 \\ 129 \\ + 24 \\ \hline 3 \end{array}$$

2. Add the tens column, including the regrouped 1.

$$\begin{array}{r} 1 \\ 40 \\ 129 \\ + 24 \\ \hline 93 \end{array}$$

3. Then add the hundreds column. Since there is only one value, write the 1 hundred in the answer.

$$\begin{array}{r} 1 \\ 40 \\ 129 \\ + 24 \\ \hline 193 \end{array}$$

You **subtract** when you want to find the **difference** between amounts. Write the greater number on top, and align the amounts on the ones column. You may also need to regroup as you subtract.

Example 2: If Sue is 57 and Kathy is 38, how many years older is Sue?

1. Find the difference in their ages. Start with the ones column. Since 7 is less than the number being subtracted (8), regroup, or **borrow**, 1 ten from the tens column. Add the regrouped amount to the ones column. Now subtract $17 - 8$ in the ones column.

$$\begin{array}{r} 417 \\ \cancel{5}7 \\ - 38 \\ \hline 9 \end{array}$$

2. Regrouping 1 ten from the tens column left 4 tens. Subtract $4 - 3$ and write the result in the tens column of your answer. Check: $19 + 38 = 57$.

$$\begin{array}{r} 417 \\ \cancel{5}7 \\ - 38 \\ \hline 19 \end{array}$$

Sue is **19 years older** than Kathy.

Example 3: Find the difference between 205 and 67.

1. Subtract. Start with the ones column. Since 5 is less than the number being subtracted (7), regroup. Since there are 0 tens in the tens column, regroup 1 hundred from the hundreds column. From 10 tens, regroup 1 ten to the ones column. Now subtract $15 - 7$ in the ones column.

$$\begin{array}{r} 9 \\ 1 \cancel{10} 15 \\ \cancel{2} \cancel{0} \cancel{5} \\ - 67 \\ \hline 8 \end{array}$$

2. Regrouping 1 ten from the tens column left 9 tens. Subtract $9 - 6$, and write the result in the tens column of your answer.

$$\begin{array}{r} 9 \\ 1 \cancel{10} 15 \\ \cancel{2} \cancel{0} \cancel{5} \\ - 67 \\ \hline 38 \end{array}$$

3. Regrouping 1 hundred from the hundreds column left 1 hundred. Subtract the hundreds column: $1 - 0$. Check: $138 + 67 = 205$.

$$\begin{array}{r} 9 \\ 1 \cancel{10} 15 \\ \cancel{2} \cancel{0} \cancel{5} \\ - 67 \\ \hline 138 \end{array}$$

Practice 2.1

A. Solve.

$$\begin{array}{r} 1. \quad 54 \\ + 23 \\ \hline \end{array}$$

$$\begin{array}{r} 3. \quad 73 \\ - 21 \\ \hline \end{array}$$

$$\begin{array}{r} 5. \quad 105 \\ + 85 \\ \hline \end{array}$$

$$\begin{array}{r} 7. \quad 100 \\ - 57 \\ \hline \end{array}$$

$$\begin{array}{r} 2. \quad 46 \\ + 54 \\ \hline \end{array}$$

$$\begin{array}{r} 4. \quad 55 \\ - 19 \\ \hline \end{array}$$

$$\begin{array}{r} 6. \quad 2386 \\ + 1692 \\ \hline \end{array}$$

$$\begin{array}{r} 8. \quad 2500 \\ - 383 \\ \hline \end{array}$$

B. Rewrite the problems in columns before solving.

$$9. 20 + 12 + 33 =$$

$$13. 144 + 238 + 101 =$$

$$17. 10,326 + 982 =$$

$$10. 245 - 131 =$$

$$14. 545 - 89 =$$

$$18. 15,890 - 705 =$$

$$11. 30 + 75 + 75 =$$

$$15. 2095 + 324 =$$

$$19. 108,755 + 22,442 =$$

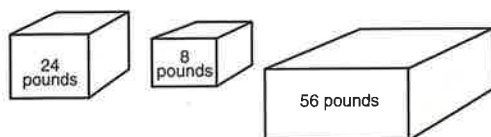
$$12. 378 - 85 =$$

$$16. 1250 - 350 =$$

$$20. 44,789 - 13,890 =$$

C. Choose the one best answer to each question.

21. What is the total weight, in pounds, of the boxes below?



- A. 78
- B. 88
- C. 150
- D. 160

22. Celia's share for lunch is \$7. If she pays with a \$20 bill, how much change should she get?

- A. \$3
- B. \$7
- C. \$13
- D. \$27

Answers and explanations begin on page 661.

Multiplication and Division

You **multiply** to combine the same amount multiple times. For example, instead of adding $24 + 24 + 24$, you could multiply 24 by 3. If a problem asks you to find the **product** of two or more numbers, you should multiply.

Example 4: Find the product of 24 and 63.

1. Align place values as you rewrite the problem in a column.
Multiply the ones place of the top number by the ones place of the bottom number: $4 \times 3 = 12$. Write the 2 in the ones place in the first partial product. Regroup the 1 ten.

$$\begin{array}{r} 1 \\ 24 \\ \times 63 \\ \hline 2 \end{array}$$

2. Multiply the tens place in the top number by 3: $2 \times 3 = 6$.
Then add the regrouped amount: $6 + 1 = 7$. Write the 7 in the tens place in the partial product.

$$\begin{array}{r} 1 \\ 24 \\ \times 63 \\ \hline 72 \end{array}$$

3. Now multiply by the tens place of 63. Write a **placeholder** 0 in the ones place in the second partial product, since you're really multiplying by 60. Then multiply the ones place of the top number by 6: $4 \times 6 = 24$. Write 4 in the partial product and regroup the 2. Multiply $2 \times 6 = 12$. Add the regrouped 2: $12 + 2 = 14$.

$$\begin{array}{r} 2 \\ 24 \\ \times 63 \\ \hline 72 \\ + 1440 \\ \hline 1512 \end{array}$$

4. Add the partial products to find the total product:
 $72 + 1440 = 1512$.

To **divide** means to find how many equal parts an amount can be divided into. The amount being divided is called the **dividend**. The number you are dividing by is the **divisor**, and the answer to a division problem is the **quotient**.

Example 5: At a garage sale, 3 children sold their old toys for a total of \$54. If they share the money equally, how much money should each child receive?

1. Divide the total amount (\$54) by the number of ways the money is to be split (3). Work from left to right. How many times does 3 go into 5? Write the answer 1 directly above the 5 in the dividend. Since $3 \times 1 = 3$, subtract $5 - 3 = 2$.

$$\begin{array}{r} 1 \\ 3 \overline{) \$54} \\ \underline{-3} \\ 2 \end{array}$$

2. Continue dividing. Bring down the 4 from the ones place in the dividend. How many times does 3 go into 24? Write the answer 8 directly above the 4 in the dividend. Since $3 \times 8 = 24$, subtract $24 - 24 = 0$.
Each child should receive \$18.

$$\begin{array}{r} 18 \\ 3 \overline{) \$54} \\ \underline{-3} \\ 24 \\ \underline{-24} \\ 0 \end{array}$$

Example 6: Divide $1006 \div 4$.

1. Divide the total amount (1006) by 4. Work from left to right. Since 4 doesn't divide into 1, use the next place value in the dividend. How many times does 4 go into 10? Write the answer 2 directly above the first 0 in the dividend. Since $4 \times 2 = 8$, subtract $10 - 8 = 2$.

$$\begin{array}{r} 2 \\ 4 \overline{) 1006} \\ \underline{-8} \\ 2 \end{array}$$

2. Continue dividing. Bring down the 0 from the tens place in the dividend. How many times does 4 go into 20? Write the answer 5 directly above the second 0 in the dividend. Since $4 \times 5 = 20$, subtract $20 - 20 = 0$. Bring down the 6 from the ones place in the dividend. How many times does 4 go into 6? Write the answer 1 above the 6 in the dividend. Since $4 \times 1 = 4$, subtract $6 - 4 = 2$. Write the **remainder** of 2 as part of the quotient.

$$\begin{array}{r} 251 \text{ r}2 \\ 4 \overline{)1006} \\ \underline{8} \\ 20 \\ \underline{20} \\ 06 \\ \underline{4} \end{array}$$

By reviewing and memorizing multiplication tables, you can save yourself precious time on the GED® *Mathematical Reasoning Test*.

Practice 2.2

A. Solve.

1. $\begin{array}{r} 121 \\ \times 4 \\ \hline \end{array}$

4. $5 \overline{)65}$

7. $\begin{array}{r} 45 \\ \times 30 \\ \hline \end{array}$

10. $10 \overline{)280}$

2. $\begin{array}{r} 250 \\ \times 4 \\ \hline \end{array}$

5. $7 \overline{)735}$

8. $\begin{array}{r} 105 \\ \times 25 \\ \hline \end{array}$

11. $15 \overline{)225}$

3. $\begin{array}{r} 342 \\ \times 8 \\ \hline \end{array}$

6. $9 \overline{)189}$

9. $\begin{array}{r} 211 \\ \times 16 \\ \hline \end{array}$

12. $19 \overline{)114}$

B. Solve. If multiplying more than two numbers, find the product of two numbers before multiplying by the next number, and so on.

13. $50 \times 5 =$

16. $1004 \div 5 =$

19. $10,760 \div 20 =$

14. $179 \div 4 =$

17. $25 \times 3 \times 2 =$

20. $12 \times 8 \times 4 =$

15. $5 \times 6 \times 10 =$

18. $7452 \times 9 =$

21. $144,140 \div 12 =$

C. Choose the one best answer to each question.

22. A fruit juice container holds 16 servings. If the serving size is 6 ounces, how many ounces does the container hold in all?

- A. 10
- B. 22
- C. 76
- D. 96

D. For question 24, write your answer on the line provided.

24. How many 2-foot lengths can be cut from the string shown below?

12 ft

Answers and explanations begin on page 661.

23. A cashier has fifteen \$5 bills. How much money does he have?

- A. \$15
- B. \$25
- C. \$75
- D. \$150

Lesson 3: GED® Test Calculator Skills

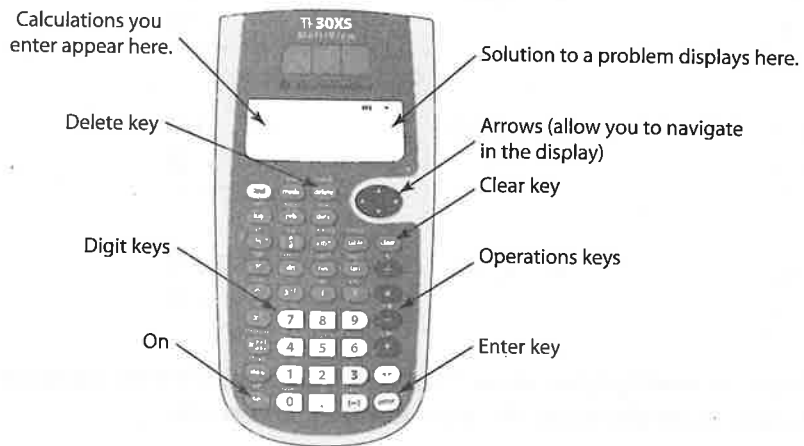
Key Ideas

- For many questions, you may use either a handheld or online version of the TI-30XS MultiView™ scientific calculator.
- Use the calculator when you think it will save you time or improve your accuracy.
- Learn the placement of the keys and how to perform specific functions before you take the test.

You will have access to an online or handheld calculator for many questions on the *Mathematical Reasoning Test* and for some questions on the *Social Studies and Science Tests*. If you wish to use a handheld calculator, you will likely need to bring your own. It must be the Texas Instruments TI-30XS MultiView™, shown here. The online version will be available when the calculator icon appears at the top of the computer screen. You will click on the icon to open the calculator and use the mouse to move it so that it does not cover the problem. See page 216 for more calculator basics.



Look at the reproduction of the calculator below and follow the examples that demonstrate how to use the calculator's operation keys.



GED® TEST TIP

You can buy a handheld TI-30XS MultiView™ calculator for practice as you work through this study guide. It may be available online, at office supply stores, or other places where calculators are sold.

Example 1: Add $63 + 97 + 58 + 32 + 81$.

1. Always clear a calculator before starting a new computation. On the TI-30XS MultiView™, use the *clear* key.

clear

2. Enter each number followed by the plus sign. As you type, the numbers and plus signs will appear on the calculator's screen. If you make a mistake, press *delete* to go back and reenter a number.

$63 + 97 + 58 + 32 + 81$

3. Press *enter* to find the total.

enter

The total, **331**, will appear on the right-hand side of the display:

331

Example 2: Find the difference between 15,789 and 9,332.

1. Always clear a calculator before starting a new computation.

clear

2. Enter the greater number first, followed by the minus operator.

15789 **=**

NOTE: Use the minus key that is on the right side of the calculator with the other operation symbols. Don't use the $(-)$ key at the bottom of the calculator; that key is used to enter a negative number.

3. Enter the number being subtracted.

9332

4. Press the *enter* key to find the answer.

enter

The answer **6457** will appear on the right-hand side of the display.

6457

Example 3: Find the product of 309 and 68.

1. Always clear a calculator before starting a new computation.

clear

2. Enter the first number, followed by the multiplication operator.

309 **×**

(The multiplication sign will appear in the display as an asterisk rather than an $\times$, but the *multiplication key* looks like an $\times$.)

3. Enter the next number.

68

4. Press the *enter* key to find the product.

enter

The answer **21012** will appear on the right-hand side of the display.

21012

Example 4: Divide 12,456 by 12.

1. Always clear a calculator before starting a new computation.

clear

2. Enter the number to be divided first, followed by the division operator.

12456 **÷**

3. Enter the number you are dividing by.

12

4. Press the *enter* key to find the quotient.

enter

The answer **1038** will appear on the right-hand side of the display.

1038

Practice 3.1

A. Practice solving the following problems on your calculator.

1. $19 + 26 + 85 + 23 =$

2. $2579 - 1392 =$

3. $4 \times 28 \times 7 =$

4. $2568 \div 107 =$

5. $12,356 + 14,728 =$

6. $107,899 - 93,457 =$

7. $209 \times 56 =$

8. $972 \div 18 =$

9. $20,540 \div 13 =$

B. Choose the one correct answer to each question.

10. Dan bought a used car with 16,741 miles on it. If the car now has 42,920 miles on it, how many miles has Dan put on the car?

A. 16,741

B. 26,179

C. 42,920

D. 59,661

11. A shipment of 20 computers arrived at a warehouse. If each computer is valued at \$995, what is the total value of the shipment?

A. \$995

B. \$1015

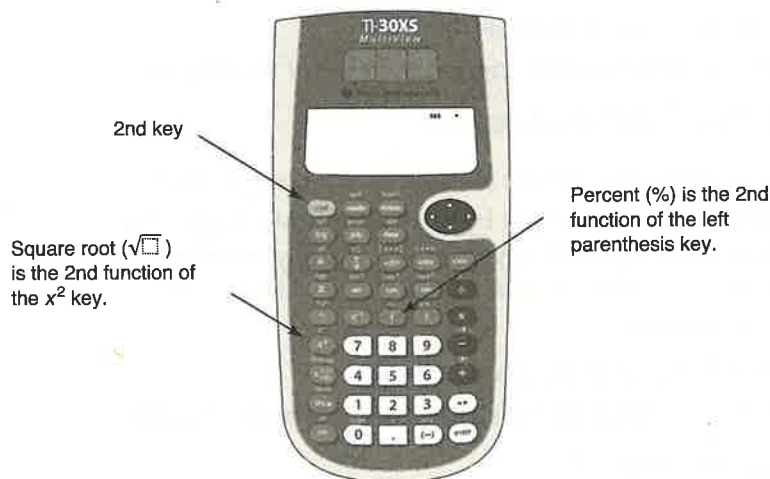
C. \$1990

D. \$19,900

Answers and explanations begin on page 661.

Using the 2nd Key for a Second Function

To access some of the functions on the TI-30XS MultiView™, you need to press the *2nd* key in the upper left corner of the keypad. This bright green key will activate the second function also shown in green above the corresponding key. To access the second function of a key, press the *2nd* key first—do not press it at the same time as the function key. Highlighted below are two commonly used second functions—square root and percent. Use the process shown here for all *2nd function* keys.



Example 5: Find the square root of 169.

1. Always clear a calculator before starting a new computation.
2. Recognize that square root is a *2nd* function over the x^2 key. Press the *2nd* key. (Note that the term *2nd* now appears in the upper left corner of the display.)
3. Next, press the x^2 key to activate the square root function in green over the key. You will see a blinking cursor under the square root.
4. Now that you have the square root function, enter the number. The number will appear under the square root symbol in the display.
5. Press the *enter* key to find the square root. The answer, 13, will appear on the right-hand side of the display.

Keys to Press

On the Display

clear

2nd

$\sqrt{\square}$

169

enter

2nd

$\sqrt{\square}$

$\sqrt{169}$

$\sqrt{169}$ 13

For more information about square roots, see page 320.

Remember: For all *2nd functions*, (1) press the *2nd* key first to activate the *2nd* function, (2) press the key, and (3) enter the numbers.

Example 6: Find the part if you are given the percent and the whole. Find 10 percent of 500.

1. Always clear a calculator before starting a new computation.
2. Enter the number you want to find the percent of.
3. Press the multiplication sign.
4. Enter the percent number.
5. Press the *2nd* key and then press the open, or left, parenthesis key to activate the percent function.
6. Press the *enter* key to find the answer.

clear
500
 $\times$
10
2nd %
enter
50

The answer, 50, will appear on the right-hand side of the display. Ten percent of 500 is 50.

Example 7: Find the percent if you are given the whole and the part. What percent of 240 is 60?

1. Always clear a calculator before starting a new computation.
2. Enter the part.
3. Press the division sign.
4. Enter the whole.
5. Press the *2nd* key and then press the close, or right, parenthesis symbol. This tells the calculator to translate the answer into a percent.
6. Press the *enter* key to find the answer.

clear
60
 $\div$
240
2nd %
enter
25%

The answer, 25%, will appear on the right-hand side of the display. Sixty is 25% of 240.

For more information about percents, see pages 264–275.

Practice 3.2

A. Practice solving the following problems on your calculator.

1. $\sqrt{625}$
2. $\sqrt{324}$
3. $\sqrt{1225}$
4. Find 20% of 680.
5. Find 10% of 1250.
6. 15 is what percent of 300?
7. Find 5% of 40.
8. Find 30% of 450.
9. 20 is what percent of 400?

B. Choose the one best answer to each question.

10. Tanya paid 20 percent of \$1680 as a down payment. How much was the down payment?
A. \$20
B. \$168
C. \$336
D. \$840
11. Aaron received a credit of \$48 on a purchase of \$960. What percent of \$960 is \$48?
A. 5%
B. 20%
C. 48%
D. 912%

Answers and explanations begin on page 661.

Lesson 4: Word Problems

Key Ideas

- Successful math problem solvers use a five-step process.
- Read carefully to understand the question and select only the information needed to answer it.
- When you have found an answer, check that it makes sense.

To pass the *Mathematical Reasoning Test*, you will need to solve word problems. These questions are easier to manage if you use a **five-step problem-solving process**. Practice this process with word problems in this book.

Step 1. What is the **question** asking me to find?

Read the problem carefully. State the question in your own words.

Step 2. What **information** from the problem do I need?

Select only the information you need to answer the question.

Step 3. Which **operation** do I need to perform—addition, subtraction, multiplication, or division?

Choose from one of the four operations above based on your understanding of the question.

Step 4. What is my **solution**?

Carry out the operation with the numbers you chose in step 2.

Step 5. Does my answer **make sense**?

Look back to make sure that you answered the question. Check that your answer makes sense.

To help you decide which operation to use, keep the following ideas in mind.

You...	in order to...
add	combine amounts of different sizes
subtract	find the difference between amounts
multiply	combine the same amount multiple times
divide	separate one amount into equal-sized groups

ON THE GED® TEST

Many Mathematical Reasoning Test questions include information that is not needed to answer the specific question. In multiple-choice questions, wrong answer choices may be based on mistakenly using this “extra” information, so read carefully.

Example 1: To cover a sofa, Sophia needs 12 yards of fabric that costs \$14 per yard. How many yards does Sophia need for 3 sofas?

- A. 4
- B. 36
- C. 42
- D. 168

The correct answer is **B. 36 yards** of fabric. The question asked you to find the total number of yards of fabric Sophia needs for 3 sofas. Note that the cost of a yard of fabric is not needed to answer the question.

Since you were asked to find the amount of fabric for 3 sofas, you should multiply 12 yards per sofa $\times$ 3 sofas = 36 yards. Here are two common errors to avoid:

- If you had mistakenly divided, you would have gotten the incorrect answer of 4 yards of fabric for 3 sofas.
- If you had used the \$14 price per yard and multiplied by either 3 sofas or 12 yards, you would have been tempted by wrong answers 42 or 168. However, the cost per yard is not needed.

Practice 4.1

1. Peter wants to repaint his 700-square-foot apartment. He calculates that he has 3500 square feet of wall space to paint. (He will not paint the floor or the ceiling.) If each gallon of paint will cover 350 square feet of wall space, how many gallons will Peter need?

A. 2
B. 5
C. 10
D. 15

2. For a family get-together, Darryl wants to be sure that each child gets 2 party favors. The party favors cost \$3 each, and there are 11 children coming. How many party favors will Darryl need?

A. 9
B. 11
C. 18
D. 22

3. Sarah and Kate live 18 miles apart, and they both work at the same office. If Sarah lives 25 miles from the office and Kate lives 30 miles from the office, how many miles farther from the office does Kate live than Sarah?

A. 5
B. 7
C. 12
D. 15

For question 4, write your answer on the line provided.

4. The Navarro family uses an average of 225 gallons of water per day, 5 gallons of which goes through the family's water filter. The Navarros' water filter can process 450 gallons before it needs to be replaced. After how many days of average water use will the family need to replace their filter?

Questions 5 and 6 are based on the following information.

Joyce owns a beauty salon, and she has posted the following information in her salon.

Service	Minutes to Complete	Price
Manicure	30	\$15
Pedicure	30	\$25
Manicure & Pedicure	45	\$35
Facial	45	\$45
Makeover	60	\$60

5. How many minutes will it take Joyce to give 3 pedicures?
- A. 30
B. 55
C. 75
D. 90
6. How much more does a customer pay for the makeover than for the manicure & pedicure combination?
- A. \$15
B. \$20
C. \$25
D. \$35

For questions 7 and 8, write your answers on the lines provided.

7. Brandon is planning his part of the local community garden. He has calculated that he can plant 6 seedlings per row, and the garden allots 7 rows to each gardener. What is the maximum number of seedlings Brandon could plant?

8. Sumayyah knits gloves and hats each winter and sells them through a local shop. If Sumayyah received \$189 from the sale of her hats and gloves this past winter, and she spent \$27 on yarn, how much money did she earn from her knitting?

\$ _____

Answers and explanations begin on page 661.

Multi-Step Word Problems

The word problems on the previous two pages involved only one operation to solve. However, on the GED® *Mathematical Reasoning Test* and other tests, you may need to do several math processes to solve a problem. These are called **multi-step problems**.

When you are working on multi-step problems, there are two important ideas to keep in mind. *What is the question asking me to find?* and *Did I answer the question?*

Review this example of a multi-step problem and the problem-solving process below.

Example 1: To win a prize, Sarah's daughter has to sell 75 boxes of cookies in 3 days. If she sold 16 boxes on day 1 and 34 boxes on day 2, how many boxes would she need to sell on day 3 to win the prize?

- (1) 18
- (2) 25
- (3) 50

Step 1. What is the question asking me to find?

The question asks me to find how many boxes Sarah's daughter needs to sell on the third day to make her goal of 75 boxes.

Step 2. What information from the problem do I need?

I know the total number of boxes she needs to sell (75 boxes).

I know how many boxes she sold on day 1 (16 boxes) and day 2 (34 boxes).

Step 3. Do I need to do only one operation, or do I need to do more than one?

I can add to find how many boxes Sarah's daughter has sold so far: $16 + 34 = 50$.

However, that doesn't tell me how many boxes she needs to sell on day 3. I need one more step.

Step 4. What is my solution?

$75 \text{ boxes total} - 50 \text{ boxes sold} = 25 \text{ boxes for day 3}$. The correct answer is (2) 25.

Step 5. Does my answer make sense? Did I answer the question?

By taking the additional step, I answered the question of how many boxes need to be sold on day 3.

This sample multi-step problem involved only two operations: adding and subtracting. On the actual test, you may need to do three or more operations to answer a question, so read and think carefully about each problem.

As you think about the example, note the most common mistake people make with multi-step problems. Often they stop short of answering the question. Notice that choice (3) 50 is a **partial solution**, but it is not the answer to the question. Using the five-step problem-solving process will help you avoid that mistake.

Practice 4.2

Questions 1–3 are based on the information below.

Farhana's produce company distributes to several restaurants. The table below shows how many cases of different produce each restaurant ordered from Farhana's company in July.

Produce Orders in July					
	Asparagus	Boston Lettuce	Carrots	Romaine Lettuce	Tomatoes
Restaurant A	2	3	1	4	3
Restaurant B	4	4	2	2	1
Restaurant C	0	0	3	4	3
Restaurant D	1	2	2	3	4
Restaurant E	3	0	3	2	1

- If Boston lettuce costs \$17 per case and romaine lettuce costs \$23 per case, how much did Restaurant D spend on lettuce ordered from Farhana in July?
A. \$85
B. \$93
C. \$103
D. \$143
- Delivery costs \$2 per case for the first 5 cases and \$1 per case for each additional case of produce. What was Restaurant B's delivery charge in July?
A. \$13
B. \$18
C. \$20
D. \$26
- If asparagus costs \$22 per case and tomatoes cost \$15 per case, which of the following restaurants spent the most on asparagus and tomatoes combined?
A. Restaurant A
B. Restaurant B
C. Restaurant C
D. Restaurant D
- At a certain store, loose-leaf paper comes only in packages of 400 sheets. If a student buys enough paper at this store to fill 3 binders with 150 sheets of paper each, how many sheets will be left over?
A. 17
B. 50
C. 350
D. 450
- Three friends are baking cupcakes for a bake sale. Each batch of 24 cupcakes requires 2 cups of flour. The friends have a single 5-pound bag of flour that contains 19 cups of flour. How many whole batches of cupcakes can they bake?
A. 9
B. 38
C. 216
D. 228
- A certain health insurance plan costs \$3000 per year for a family of six. If each member of the family has \$750 in medical expenses in a year, and the plan pays 100 percent of those expenses, how much will the family save by purchasing the plan?
A. \$1500
B. \$2000
C. \$2500
D. \$3000

Answers and explanations begin on page 661.

Lesson 5: Distance and Cost

Key Ideas

- The following all indicate multiplication:

$$n \times r \quad nr \quad n(r) \quad n \cdot r$$

- The following indicate division:

$$c \div n \quad \frac{c}{n}$$

- A formula can be rewritten to solve for each of its variables.

$$c = nr \quad \frac{c}{n} = r \quad \frac{c}{r} = n$$

$$d = rt \quad \frac{d}{r} = t \quad \frac{d}{t} = r$$

ON THE GED® TEST

Substitute all of the values for variables back into a formula to check your answer.

Distance

On the *Mathematical Reasoning Test*, you'll see questions that will require you to apply formulas. When a relationship is constant, use a formula to set up the information about how the different parts relate to each other. One of the most commonly tested formulas deals with distance, rate, and time. Distance is a product of the rate at which something travels and the amount of travel time:

$$\text{Distance} = \text{rate} \times \text{time}, \quad \text{or} \quad d = rt$$

Notice that letters, or variables, can be used to represent the different parts of a formula. A formula allows you to substitute known values for certain variables and solve for the unknown variable.

Example 1: How many miles can you travel if you drive at an average speed of 55 miles per hour for 3 hours?

- You know the rate (55 miles per hour) and the time (3 hours). Substitute the values in the distance formula. $\text{distance} = 55 \times 3$

- Multiply to find the distance. $55 \times 3 = 165 \text{ miles}$

Cost

Another helpful formula is the cost formula. It expresses the relationship between cost, the number of units, and rate (price per unit). **NOTE:** The word *per* means for every one unit. The cost formula can be written as follows:

$$\text{Total cost} = (\text{number of units}) \times (\text{price per unit}), \quad \text{or} \quad c = nr$$

Example 2: At a bakery, a package of frosted cookies is priced at \$3 per package. If a teacher treats her class by buying 4 packages, how much would the cookies cost before tax?

- You know the number of units (4 packages) and the price per unit (\$3 per package). Substitute the values in the cost formula. $c = nr$
 $\text{total cost} = 4 \times \3

- Multiply to find the total cost. $4 \times \$3 = \12

If you know any two of the three variables in a formula, you can solve for the third variable.

Example 3: Max bought a set of 4 floor mats for \$44. How much was the price per floor mat?

- You know the total cost (\$44) and the number of units (4 floor mats). Rewrite the formula to solve for the price per unit (r). $c = nr$
 $r = \frac{c}{n}$
- Substitute the known values in the cost formula. Divide to find the price per unit. $\frac{\$44}{4} = \11

Practice 5

- A. Each problem below includes two of the three variables from either the distance formula or the cost formula. Write the missing variable you need to solve for. Then decide which of the following formula variations you would use in each situation. The first one is done for you.

$$d = rt \quad \frac{d}{r} = t \quad \frac{d}{t} = r \quad c = nr \quad \frac{c}{n} = r \quad \frac{c}{r} = n$$

- | | |
|--|--|
| 1. Given: distance and time
Solve for: rate
Formula: $\frac{d}{t} = r$ | 4. Given: cost and number of units
Solve for: _____
Formula: _____ |
| 2. Given: rate and time
Solve for: _____
Formula: _____ | 5. Given: number of units and price per unit
Solve for: _____
Formula: _____ |
| 3. Given: distance and rate
Solve for: _____
Formula: _____ | 6. Given: cost and price per unit
Solve for: _____
Formula: _____ |

- B. Use the formulas provided in part A above to help you set up the problems. Solve for the unknown variable.

- | | |
|--|--|
| 7. Find the total cost of 4 flats of plants at \$12 each. | 12. Find the distance traveled by a car averaging 60 miles per hour for 3 hours. |
| 8. Find the total cost of 12 boxes of cookies if each box costs \$3. | 13. Find the distance traveled by a train averaging 50 miles per hour for 4 hours. |
| 9. If 4 tires cost \$320, how much does a single tire cost? | 14. How long does it take for a bus to travel 25 miles at an average rate of 25 miles per hour? |
| 10. How many tickets would you get for \$25 if raffle tickets cost \$5 apiece? | 15. If a train travels 270 miles in 3 hours, what is the train's average speed? |
| 11. If you paid \$20 for 10 bus transfer tickets, how much did you pay per ticket? | 16. How long does it take to complete a delivery route of 75 miles at a rate of 25 miles per hour? |

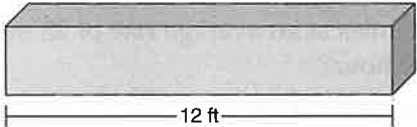
- C. Choose the one best answer to each question.

- | | |
|--|---|
| 17. A company sold a total of \$640 in gift boxes. If the gift boxes cost \$20 apiece, how many gift boxes did the company sell?
A. 32
B. 320
C. 660
D. 1280 | 18. A truck driver traveled 275 miles in 5 hours. What was his average speed in miles per hour?
A. 1375
B. 280
C. 270
D. 55 |
|--|---|

Answers and explanations begin on page 661.

Number Sense and Problem Solving Practice Questions

Directions: You MAY use your calculator.

1. A beverage container holds 12 servings. If the serving size is 8 ounces, how many ounces does the container hold in all?
A. 20
B. 32
C. 48
D. 96
2. Sales at 3 concession stands are \$839, \$527, and \$726. What is the total amount in sales?
A. \$1581
B. \$2092
C. \$2178
D. \$2517
3. If you want to cut 24 two-foot braces, how many boards of the length shown below would you need?

A. 4
B. 6
C. 12
D. 24
4. Using your calculator, find the value of $\sqrt{441}$.
A. 11
B. 21
C. 221
D. 441
5. Angelo bought a used car with 39,451 miles on it. If the car now has 70,040 miles on it, how many miles has Angelo driven the car?
A. 30,589
B. 39,459
C. 70,040
D. 109,491
6. Use your calculator to solve this problem. If Emory paid 20 percent of \$3280 as a down payment, how much was the down payment?
A. \$164
B. \$328
C. \$656
D. \$6560
7. Inventory shows that a warehouse has 45 printers in stock. If each printer is valued at \$125, what is the total value of the printer inventory?
A. \$5625
B. \$170
C. \$80
D. \$45
8. Lydia can drive 180 miles in 3 hours. On Tuesday, she drove for 7 hours at that rate. How many miles did she drive on Tuesday?
A. 60
B. 420
C. 600
D. 1260

9. Janelle wants to drive from Danville to Brownsville. If she averages 60 miles per hour, how many hours will it take her to drive the distance?



- A. 6
B. 8
C. 60
D. 540
10. A company sold a total of \$1440 in gift bears for Valentine's Day. If the gift bears cost \$15 apiece, how many gift bears did it sell?
- A. 15
B. 96
C. 144
D. 1440
11. In addition to interest charges, Richard's credit card company charges a \$25 late fee for payments made after the payment due date. If he was charged a late fee for 8 monthly bills, how much could he have saved by paying the bills on time?
- A. \$200
B. \$80
C. \$33
D. \$25
12. A waiter has seven \$5 bills and eighteen \$1 bills from tips. In all, how much does he have in tips?
- A. \$18
B. \$25
C. \$35
D. \$53

13. April has taken her car in for the recommended oil and filter change every 3,500 miles. If April bought her car brand-new, and the odometer now shows just over 17,500 miles, how many oil changes has her car received?

- A. 5
B. 14
C. 123
D. 1236

14. A clinic treated 536 children over a 4-month period. At this rate, how many children did the clinic treat in 1 month?

- A. 134
B. 536
C. 540
D. 2144

15. Attendance at a local play was 438 Friday night, 820 Saturday night, and 636 Sunday afternoon. How many more people attended the play on Sunday than on Friday?

Write your answer on the line below.

16. Raquel has 4 payments left on her car. If each payment is \$268, how much does she still owe on her car?

Write your answer on the line below.

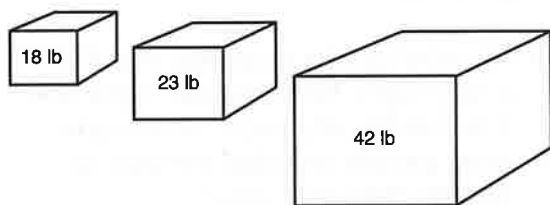
17. In what order should items weighing 51 pounds, 40 pounds, 48 pounds, and 44 pounds be stacked if you want them in order from heaviest to lightest?

A. 51, 44, 40, 48
B. 40, 44, 48, 51
C. 51, 48, 44, 40
D. 51, 40, 44, 48

18. Which of the following correctly shows 2,354,769 rounded to the nearest ten thousand?

A. 2,400,000
B. 2,355,000
C. 2,350,000
D. 2,000,000

19. What is the total weight, in pounds, of the packages below?



- A. 83
B. 65
C. 60
D. 42
20. Jason paid a \$14 dinner bill with a \$20 bill. How much change should he receive?
- A. \$6
B. \$7
C. \$14
D. \$34

21. Maria spent 8 minutes installing a new showerhead, 33 minutes rodding out a drain, and 18 minutes fixing a leaking faucet. About how many minutes did it take Maria to complete all three tasks?

A. 90
B. 60
C. 30
D. 20

22. Carla drove 248 miles in 4 hours. What was her average rate of speed, in miles per hour?

Write your answer on the line below.

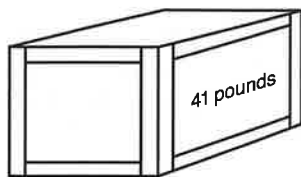
23. A bulk bag of nuts contains 144 ounces of nuts. If the nuts are packaged in smaller 8-ounce bags, how many bags will there be?

A. 8
B. 12
C. 18
D. 136

24. If you drove 299 miles on 9 gallons of gasoline, about how many miles per gallon did the car get?

A. 10
B. 30
C. 270
D. 300

25. A shipment of 33 crates like the one shown below is delivered. Approximately how many pounds did workers unload?



- A. 40
B. 120
C. 1200
D. 2000
26. Four friends bought a birthday cake for \$21 and balloons for \$15. If they divided the cost equally, how much did each friend pay toward the birthday party?
- A. \$4
B. \$5
C. \$9
D. \$36
27. David paid \$20 toward a dinner bill of \$128. If the remainder of the bill is divided equally among the remaining 9 people in the group, how much should each person other than David pay?
- A. \$12
B. \$18
C. \$20
D. \$108
28. A driver traveled 4 hours at an average rate of 65 miles per hour. How many miles did the person drive?
- A. 61
B. 69
C. 240
D. 260

29. Using the following information, how much would a large pizza with 3 toppings cost?

Large 1-Topping Pizza for \$14 \$2 for Each Additional Topping

- A. \$20
B. \$18
C. \$16
D. \$14
30. Bagels are 2 for \$1. What is the maximum number of bagels you could buy for \$7?
- A. 7
B. 10
C. 14
D. 15
31. How many months would it take to save \$1800 at \$75 per month?
- Write your answer on the line below.
- _____
32. How many 45-page documents would a binder hold if its maximum capacity is 630 sheets of paper?
- Write your answer on the line below.
- _____

Answers and explanations begin on page 662.

Decimals and Fractions

Lesson 1: Decimal Basics

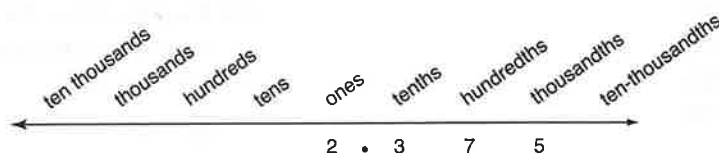
Key Ideas

- The decimal point separates the whole number from the decimal part.
- To round decimals, use the same rules as with rounding whole numbers.
- Adding zeros to the right of the last digit after the decimal does not change the number's value.

The Decimal System

Decimals are numbers that use place value to show amounts less than 1. You already use decimals when working with money. For example, in the amount \$10.25, you know that the digits to the right of the **decimal point** represent cents, or hundredths of a dollar.

The first four decimal place values are labeled on the chart below.



The number 2.375 is shown on the chart. Read *and* in place of the decimal point. After reading the decimal part, say the place value of the last decimal digit. This number would be read "two *and* three hundred seventy-five *thousandths*."

Rounding

Round decimals the same way you round whole numbers.

Example 1: A calculator display reads 3.62835. Round to the hundredths place.

1. Look at the digit to the right of the hundredths place. 3.62835
2. If the digit to the right is 5 or greater, round up. If the digit is less than 5, don't change the number. Then drop all digits to the right of the place you are rounding to. Since 8 is greater than 5, round up.
3.62835 rounds to
3.63.

Comparing and Ordering

Comparing decimals is similar to comparing whole numbers.

Example 2: Matt ran the 400-meter race in 45.8 seconds. Alonzo ran the same race in 45.66 seconds. Which runner had the faster time?

1. Line up the decimal points. Add a zero at the **end** of 45.8 so that both times have the same number of digits after the decimal. 45.80
45.66
2. Compare the decimal parts of the numbers as though they were whole numbers. **Alonzo's time was faster.** 80 is greater than 66, so 45.8 is greater than 45.66.

ON THE GED® TEST

The Mathematical Reasoning Test may show a zero written before the decimal point. The zero does not change the value of the number.
Example: 0.75 = .75

When you compare more than two numbers, it is helpful to compare one place-value column at a time, working from left to right.

Example 3: Arrange the numbers 0.85, 1.8, 0.8, and 0.819 in order from greatest to least.

- | | |
|---|---------------------------------------|
| 1. Write the numbers in a column, lining up the decimal points.
Add zeros so that the numbers have the same number of decimal places. | 0.850
1.800
0.800
0.819 |
| 2. Compare the digits, working from left to right. Only 1.8 has a whole number part, so it is greatest. The remaining numbers each have 8 in the tenths column. Looking at the hundredths column, 0.85 is next, followed by 0.819. The least number is 0.8. | In order: 1.8
0.85
0.819
0.8 |

Practice 1

A. Round these numbers as directed.

- | | |
|---|--|
| 1. Round 3.75 to the tenths place. | 4. Round 0.66667 to the nearest thousandth. |
| 2. Round 5.908 to the ones place. | 5. Round 8.125 to the nearest tenth. |
| 3. A calculator display reads 0.4285714.
Round to the nearest hundredth. | 6. A calculator display reads 2.7142857.
Round to the nearest thousandth. |

B. In each of the following pairs, which number is greater?

- | | | |
|------------------|------------------|---------------------|
| 7. 0.45 or 0.449 | 9. 4.68 or 4.086 | 11. 1.0275 or 1.029 |
| 8. 0.008 or 0.08 | 10. 0.75 or 1.85 | 12. 0.14 or 0.104 |

C. Write these numbers in order from least to greatest.

- | | |
|---------------------------|----------------------------|
| 13. 5.6 5.08 5.8 5.802 | 15. 14.005 4.52 4.8 4.667 |
| 14. 0.1136 0.12 0.2 0.115 | 16. 0.8023 0.8 0.803 0.823 |

D. Choose the one best answer to each question.

- | | |
|---|---|
| 17. In a circuit board assembly, the weights of three parts are 0.572 grams, 0.0785 grams, and 0.6 grams. Which of the following lists the weights in order from <u>greatest to least</u> ? | 18. Which of the following correctly shows 1.3815 rounded to the nearest hundredth? |
| A. 0.0785 g, 0.572 g, 0.6 g | A. 1.4 |
| B. 0.6 g, 0.0785 g, 0.572 g | B. 1.382 |
| C. 0.6 g, 0.572 g, 0.0785 g | C. 1.381 |
| D. 0.572 g, 0.6 g, 0.0785 g | D. 1.38 |

Answers and explanations begin on page 662.

Lesson 2: Decimal Operations

Key Ideas

- Always add and subtract like place-value columns.
- Always line up the decimal points when you write a problem in columns.
- When adding or subtracting, place the decimal point in the answer directly below the decimal point in the problem.

Addition and Subtraction

Adding decimals is much like adding whole numbers. The trick is to make sure you have lined up the place-value columns correctly. You can do this by writing the numbers in a column and carefully lining up the decimal points.

Example 1: Add $0.37 + 13.5 + 2.638$.

1. Write the numbers in a column, lining up the decimal points.
2. You may add placeholder zeros so that the decimals have the same number of decimal places.
3. Add. Start on the right and add each column. Re-group, or carry, as you would with whole numbers.
4. Place the decimal point in the answer directly below the decimal points in the problem.

$$\begin{array}{r} 0.370 \\ 13.500 \\ + 2.638 \\ \hline 16.508 \end{array}$$

To find the difference between decimals, write the numbers in a column with the greater number on top. Make sure the decimal points are in a line.

Example 2: Find the difference between 14.512 and 8.7.

1. Write the numbers in a column, lining up the decimal points. Add placeholder zeros so that the numbers have the same number of decimal places.
2. Subtract. Regroup, or borrow, as needed. Place the decimal point in the answer directly in line with the decimal points in the problem.

$$\begin{array}{r} 14.512 \\ - 8.700 \\ \hline 5.812 \end{array}$$

GED® TEST TIP

To make sure that your answer makes sense, mentally round the numbers to the nearest whole number and then add or subtract. The result should be close to your answer.

The greater number may have fewer or no decimal places. In the next example, a decimal is subtracted from a whole number.

Example 3: What does 9 minus 3.604 equal?

1. Line up the place-value columns. Put a decimal point after the whole number 9 and add placeholder zeros.
2. Subtract, regrouping as needed. Place the decimal point in the answer.

$$\begin{array}{r} 9.000 \\ - 3.604 \\ \hline 5.396 \end{array}$$

Practice 2.1

A. Solve. You MAY NOT use a calculator.

$$\begin{array}{r} 1. \quad 4.025 \\ + 3.971 \\ \hline 7.996 \end{array}$$

$$\begin{array}{r} 2. \quad 6.5 \\ + 4.008 \\ \hline \end{array}$$

$$\begin{array}{r} 3. \quad 2.80 \\ + 9.46 \\ \hline 12.26 \end{array}$$

$$\begin{array}{r} 4. \quad 8.04 \\ - 2.19 \\ \hline 5.85 \end{array}$$

$$\begin{array}{r} 5. \quad 8.5 \\ - 1.074 \\ \hline \end{array}$$

$$\begin{array}{r} 6. \quad 10.00 \\ - 7.89 \\ \hline 2.11 \end{array}$$

$$\begin{array}{r} 7. \quad 17.294 \\ + 0.8 \\ \hline \end{array}$$

$$\begin{array}{r} 8. \quad 4.07 \\ + 1.047 \\ \hline \end{array}$$

$$\begin{array}{r} 9. \quad 17.52 \\ + 3.8 \\ \hline \end{array}$$

$$\begin{array}{r} 10. \quad 3.8 \\ - 2.905 \\ \hline \end{array}$$

$$\begin{array}{r} 11. \quad 14.64 \\ - 10.8 \\ \hline \end{array}$$

$$\begin{array}{r} 12. \quad 100.5 \\ - 98.15 \\ \hline \end{array}$$

$$\begin{array}{r} 21. \quad 0.010 \\ + 2.052 \\ + 0.960 \\ + 1.500 \\ \hline 4.522 \end{array}$$

$$\begin{array}{r} 24. \quad 32.880 \\ - 15.675 \\ \hline 17.225 \end{array}$$

13. $0.236 + 2.4 + 2.87 =$

17. $1.02 - 0.87 =$

21. $0.01 + 2.052 + 0.96 + 1.5 =$

14. $38.06 - 16.9 =$

18. $0.45 + 1.8 + 0.07 + 2.56 =$

22. $12.9 - 10.54 =$

15. $0.006 + 0.09 + 0.549 =$

19. $12.5 - 0.7 =$

23. $0.68 + 12.3 + 4.9 =$

16. $8.5 - 6.074 =$

20. $25 - 10.984 =$

24. $32.9 - 15.675 =$

B. Choose the one best answer to each question.

25. James ran 3 miles. His times for the individual miles were 7.2 minutes, 6.8 minutes, and 8.25 minutes. How long did it take him, in minutes, to run the 3-mile distance?

A. 22.25
B. 22.7
C. 23.35
D. 96.5

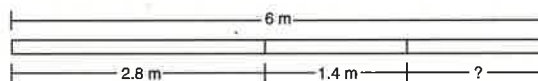
26. Claudia earns overtime pay when she works more than 40 hours in one week. How many hours of overtime pay did she work for the week of March 4?

Work Record for March 4-10

March 4	8.5
March 5	Off
March 6	9.25
March 7	8.75
March 8	10
March 9	Off
March 10	7.75

A. 44.25
B. 40.0
C. 4.25
D. 2.25

27. A plumber cut two lengths of pipe measuring 2.8 and 1.4 meters from a 6-meter length.



Assuming there was no waste when the cuts were made, what is the length in meters of the remaining piece?

A. 1.8
B. 3.2
C. 4.2
D. 7.4

$$\begin{array}{r} 2.8 \\ + 1.4 \\ \hline 4.2 \end{array}$$

$$\begin{array}{r} 6.0 \\ - 4.2 \\ \hline 1.8 \end{array}$$

28. Mona purchased the following art supplies: a storage box for \$16.98, a set of art markers for \$31.78, and a pad of paper for \$6.50. What was the cost of the three items?

A. \$48.76
B. \$53.26
C. \$55.26
D. \$61.76

$$\begin{array}{r} 16.98 \\ + 31.78 \\ + 6.50 \\ \hline 55.26 \end{array}$$

Answers and explanations begin on page 662.

Multiplication and Division

The rules you used to multiply whole numbers can be used to multiply decimals. You don't have to line up the decimal points. You will wait until you are finished multiplying before you place the decimal point in the answer. The number of decimal places in the answer equals the total number of decimal places in the numbers you are multiplying.

Example 4: Find the product of 2.6 and 0.45.

1. Set up the problem as though you were multiplying the whole numbers 26 and 45.

$$\begin{array}{r} 2.6 \\ \times .45 \\ \hline \end{array}$$

2. Ignore the decimal points while you multiply.

$$\begin{array}{r} 2.6 \\ \times .45 \\ \hline \end{array}$$

3. Now count the decimal places in the numbers you multiplied. The number 2.6 has one decimal place, and 0.45 has two decimal places, for a total of three.

$$\begin{array}{r} 130 \\ 1040 \\ \hline 1.170 \\ \hline \end{array}$$

4. Starting from the right, count three places to the left and insert the decimal point. Thus, the answer is 1.17.

When you divide decimals, you must figure out where the decimal point will go in the answer before you divide.

Example 5: Divide 14.4 by 6.

1. Set up the problem. Since the divisor (the number you are dividing by) is a whole number, place the decimal point in the answer directly above the decimal point in the dividend (the number you are dividing).

$$\begin{array}{r} 2.4 \\ 6 \overline{)14.4} \\ \underline{-12} \\ 24 \\ \underline{-24} \\ 0 \end{array}$$

2. Divide. Use the rules you learned for dividing whole numbers. The answer is 2.4.

If the divisor is a decimal, you must move the decimal points in both the divisor and the dividend before you divide.

Example 6: Divide 4.9 by 0.35.

$$\begin{array}{r} .35 \overline{)4.90} \\ \hline \end{array}$$

1. Set up the problem. There are two decimal places in the divisor. Move the decimal point in *both* the divisor and the dividend two places to the right. Note that you need to add a zero in the dividend in order to move the decimal two places.

$$\begin{array}{r} 14. \\ 35 \overline{)49.0} \\ \underline{-35} \\ 140 \\ \underline{-140} \\ 0 \end{array}$$

2. Place the decimal point in the quotient directly above the decimal point in the dividend.

3. Divide. The correct answer is 14.

NOTE: You may not need to finish dividing in order to choose the correct answer. You may be able to eliminate all but one of the answer choices after only one or two division steps.

Practice 2.2

A. Solve. You MAY NOT use a calculator.

$$\begin{array}{r} 1. \quad 5.3 \\ \times 0.5 \\ \hline \end{array}$$

$$4. \quad 6 \overline{)3.12}$$

$$\begin{array}{r} 7. \quad 6.25 \\ \times 1.4 \\ \hline \end{array}$$

$$10. \quad 1.25 \overline{)30}$$

$$\begin{array}{r} 2. \quad 64 \\ \times 0.2 \\ \hline \end{array}$$

$$5. \quad 8 \overline{)28.8}$$

$$\begin{array}{r} 8. \quad 13.5 \\ \times 0.25 \\ \hline \end{array}$$

$$11. \quad 2.8 \overline{)39.76}$$

$$\begin{array}{r} 3. \quad 12.4 \\ \times 0.04 \\ \hline \end{array}$$

$$6. \quad 5 \overline{)20.45}$$

$$\begin{array}{r} 9. \quad 9.62 \\ \times 1.005 \\ \hline \end{array}$$

$$12. \quad 0.003 \overline{)47.4}$$

$$13. \quad 15.5 \times 2.2 =$$

$$16. \quad 1.32 \div 0.5 =$$

$$19. \quad 3.36 \times 1.1 =$$

$$14. \quad 0.944 \div 0.4 =$$

$$17. \quad 2.75 \times 0.6 =$$

$$20. \quad 15.03 \div 15 =$$

$$15. \quad 2.05 \times 0.32 =$$

$$18. \quad 12.825 \div 3 =$$

$$21. \quad 0.12 \times 0.06 =$$

B. Choose the one best answer to each question.

22. One container of floor cleaner holds 3.79 liters. If Zachary bought 4 containers, how many liters of cleaner did he buy?

A. 0.9475
B. 7.79
C. 12.83
D. 15.16

23. Ribbon costs \$0.45 per foot. A sewing project calls for 20.5 feet of ribbon. To the nearest cent, what will be the cost of the ribbon for the project?

A. \$0.92
B. \$9.23
C. \$9.90
D. \$45.56

24. Armando drove 278.7 miles over a 3-day period. On average, how many miles did he drive each day?

A. 9.3
B. 90.3
C. 92.9
D. 836.1

Questions 25 and 26 are based on the following information.

Cereal	Net Weight	Servings per Box
Toasted Oats	22.8 oz	19
Crisp Rice	16.9 oz	13
Honey Mix	12.5 oz	10

25. A box of Toasted Oats cereal is priced at \$4.94. What is the cost per serving? (Hint: Divide the price by the number of servings.)

A. \$0.49
B. \$0.29
C. \$0.26
D. \$0.22

26. Lee bought 4 boxes of Honey Mix cereal. How many ounces of cereal did she buy?

A. 31.25
B. 50.0
C. 67.6
D. 91.2

Answers and explanations begin on page 662.

Lesson 3: Fraction Basics

Key Ideas

- The bottom number of a fraction tells how many parts the group or object has.
- The top number tells how many parts you are working with.
- When the top number is greater than the bottom number, the fraction is greater than 1.

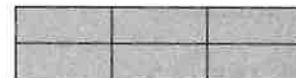
A **fraction** uses two numbers to represent part of a whole. The bottom number, called the **denominator**, tells how many equal parts are in the whole group or item. The top number, called the **numerator**, tells how many parts you are working with.

There are 4 equal parts in this rectangle. Since 3 are shaded, we say that $\frac{3}{4}$ of the rectangle is shaded.

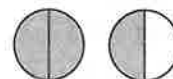


In a proper fraction, the numerator is less than the denominator. A **proper fraction** represents a quantity less than 1. An **improper fraction** is equal to or greater than 1.

There are 6 equal parts in the figure, and 6 are shaded; therefore, $\frac{6}{6}$ of the figure is shaded. $\frac{6}{6} = 1$



In this grouping, each figure is divided into 2 equal parts. A total of 3 parts are shaded, so $\frac{3}{2}$ are shaded.



A **mixed number** is another way to show an amount greater than 1. It consists of a whole number and a proper fraction. Another name for the shaded portion in the last figure is $1\frac{1}{2}$. The improper fraction $\frac{3}{2}$ equals $1\frac{1}{2}$.

You can also change an improper fraction to a whole or mixed number.

Example 1: Change $\frac{16}{5}$ to a mixed number.

1. Divide the numerator (16) by the denominator (5). $16 \div 5 = 3 \text{ r}1$.
Since 16 is not evenly divisible by 5, there is a remainder of 1.
2. The answer becomes the whole number, and the remainder becomes the numerator of the proper fraction. The denominator is the same as the original fraction.

$$\frac{16}{5} = 3\frac{1}{5}$$

You can also change a mixed number to an improper fraction.

Example 2: Change $7\frac{2}{3}$ to an improper fraction.

1. Multiply the whole number (7) by the denominator of the fraction (3), and add the numerator (2).
 $7 \times 3 = 21$
 $21 + 2 = 23$
2. Write the sum over the denominator of the original fraction.

$$7\frac{2}{3} = \frac{23}{3}$$

GED® TEST TIP

If you have trouble visualizing the fractions in a problem, draw a quick sketch of the information presented in the problem similar to the ones on this page.

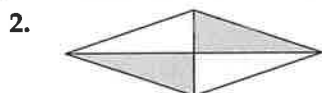
To perform operations with fractions, you need to be able to write equal fractions in higher or lower terms. The **terms** are the numerator and the denominator. A fraction is **reduced to lowest terms** when the two terms do not have any common factor except 1.

To **raise** a fraction, multiply both terms by the same number: $\frac{3}{4} = \frac{3 \times 3}{4 \times 3} = \frac{9}{12}$.

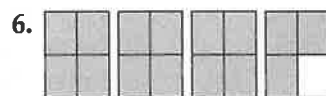
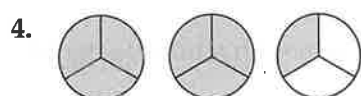
To **reduce** a fraction, divide both terms by the same number: $\frac{10}{15} = \frac{10 \div 5}{15 \div 5} = \frac{2}{3}$.

Practice 3

A. Write a proper fraction for the shaded portion of each figure.



B. Write an improper fraction and a mixed number for the shaded portion of each figure.



C. Write improper fractions as mixed numbers and mixed numbers as improper fractions.

7. $\frac{17}{3} =$

9. $\frac{24}{6} =$

11. $\frac{19}{4} =$

13. $\frac{43}{9} =$

15. $\frac{33}{4} =$

8. $3\frac{3}{5} =$

10. $5\frac{2}{9} =$

12. $2\frac{5}{12} =$

14. $1\frac{3}{4} =$

16. $5\frac{7}{10} =$

D. Write an equal fraction with the given denominator.

17. $\frac{3}{4} = \frac{\quad}{16}$

18. $\frac{1}{3} = \frac{\quad}{21}$

19. $\frac{4}{5} = \frac{\quad}{60}$

20. $\frac{3}{8} = \frac{\quad}{40}$

21. $\frac{6}{25} = \frac{\quad}{100}$

(Hint for question 17: $4 \times ? = 16$)

E. Reduce each fraction to lowest terms.

22. $\frac{21}{28} =$

23. $\frac{4}{24} =$

24. $\frac{12}{20} =$

25. $\frac{26}{30} =$

26. $\frac{60}{90} =$

F. Choose the one best answer to each question.

27. Eighteen out of every 24 people surveyed say they went to at least one movie in December. What fraction of the people surveyed went to at least one movie in December?

A. $\frac{3}{4}$

B. $\frac{2}{3}$

C. $\frac{1}{3}$

D. $\frac{1}{4}$

28. Which of the following fractions equals $\frac{2}{5}$?

A. $\frac{15}{100}$

B. $\frac{30}{100}$

C. $\frac{40}{100}$

D. $\frac{80}{100}$

Answers and explanations begin on page 663.

Lesson 4: Fraction Operations

Addition and Subtraction

You can add or subtract **like fractions**. Like fractions have a **common denominator**. In other words, their denominators are the same.

Key Ideas

- You can add or subtract fractions, but they must have the same denominator.
- If denominators are not the same, find a common denominator and raise one or both fractions.
- Always reduce answers to lowest terms and change improper fractions to mixed numbers.

Example 1: Add $\frac{3}{10} + \frac{5}{10}$.

1. Since the denominators are the same, add the numerators. $\frac{3}{10} + \frac{5}{10} = \frac{8}{10}$

2. Reduce the answer to lowest terms. $\frac{8}{10} = \frac{8 \div 2}{10 \div 2} = \frac{4}{5}$

Example 2: Subtract $\frac{2}{9}$ from $\frac{7}{9}$.

Subtract the numerators. The answer is already in lowest terms. $\frac{7}{9} - \frac{2}{9} = \frac{5}{9}$

If the denominators are not the same, raise one or both fractions to higher terms so that they become like fractions.

Example 3: Add $\frac{5}{6} + \frac{1}{4}$.

1. One way to find a common denominator is to think of the multiples of both denominators. The lowest is 12.
Multiples of 6: 6, 12, 18
of 4: 4, 8, 12, 16

2. Raise each fraction to higher terms with a denominator of 12. $\frac{5 \times 2}{6 \times 2} = \frac{10}{12}$, $\frac{1 \times 3}{4 \times 3} = \frac{3}{12}$

3. Add the like fractions. Rewrite the sum as a mixed number. $\frac{10}{12} + \frac{3}{12} = \frac{13}{12} = 1\frac{1}{12}$

GED® TEST TIP

If your solution to a fraction problem is not one of the given answer choices, make sure that you reduced your answer to lowest terms. On the Mathematical Reasoning Test, answer choices are always written in lowest terms.

Use the same process to add or subtract mixed numbers. Example 4 shows how to regroup when subtracting mixed numbers.

Example 4: Subtract $4\frac{1}{16} - 1\frac{3}{8}$.

1. Raise the second fraction so that it also has a denominator of 16. $\frac{3 \times 2}{8 \times 2} = \frac{6}{16}$

2. Set up the problem. To subtract the fractions, you need to regroup 1 from the whole number column and add it to the top fraction. $4\frac{1}{16} = 3\frac{16}{16} + \frac{1}{16} = 3\frac{17}{16}$

3. Subtract the fractions and then the whole numbers. $3\frac{17}{16} - 1\frac{6}{16} = 2\frac{11}{16}$

Practice 4.1

A. Solve. Reduce answers to lowest terms. Simplify improper fractions as mixed numbers.

1.
$$\begin{array}{r} \frac{3}{8} \\ + \frac{1}{8} \\ \hline \end{array}$$

3.
$$\begin{array}{r} \frac{8}{9} \\ - \frac{5}{9} \\ \hline \end{array}$$

5.
$$\begin{array}{r} \frac{1}{4} \\ + \frac{2}{3} \\ \hline \end{array}$$

7.
$$\begin{array}{r} \frac{9}{10} \\ - \frac{3}{5} \\ \hline \end{array}$$

9.
$$\begin{array}{r} 2\frac{1}{5} \\ + 1\frac{2}{3} \\ \hline \end{array}$$

2.
$$\begin{array}{r} \frac{1}{6} \\ + \frac{5}{6} \\ \hline \end{array}$$

4.
$$\begin{array}{r} \frac{7}{12} \\ - \frac{5}{12} \\ \hline \end{array}$$

6.
$$\begin{array}{r} \frac{1}{2} \\ + \frac{5}{8} \\ \hline \end{array}$$

8.
$$\begin{array}{r} \frac{7}{9} \\ - \frac{1}{2} \\ \hline \end{array}$$

10.
$$\begin{array}{r} 4\frac{1}{2} \\ - 2\frac{3}{4} \\ \hline \end{array}$$

11. $5\frac{5}{6} + 2\frac{2}{3} =$
 12. $6\frac{7}{8} + 4\frac{3}{4} =$
 13. $12\frac{1}{10} + 9\frac{3}{5} =$
 14. $2\frac{2}{9} + \frac{2}{3} + 4\frac{5}{6} =$
 15. $3\frac{1}{3} + 5\frac{2}{3} + 3\frac{5}{6} =$

16. $\frac{3}{8} + \frac{7}{12} + 1\frac{2}{3} =$
 17. $16\frac{2}{3} + 25\frac{3}{4} =$
 18. $10\frac{1}{2} + 8\frac{4}{5} + 3\frac{1}{4} =$
 19. $8\frac{1}{2} - 3\frac{4}{9} =$
 20. $15 - 3\frac{7}{8} =$

21. $14\frac{1}{4} - 10\frac{3}{7} =$
 22. $9\frac{11}{12} - 8\frac{5}{8} =$
 23. $6 - 3\frac{4}{7} =$
 24. $13\frac{1}{3} - 4\frac{4}{9} =$
 25. $5\frac{5}{7} - 4\frac{4}{5} =$

B. Choose the one best answer to each question.

26. To make the top of a dining room table, Craig glues a piece of oak that is $\frac{5}{16}$ inch thick to a piece of pine that is $\frac{7}{8}$ inch thick. What is the total thickness, in inches, of the tabletop?

- A. $\frac{9}{16}$
 B. $1\frac{3}{16}$
 C. $1\frac{1}{4}$
 D. $1\frac{9}{16}$

27. Carol will use the two bolts shown below to assemble a book cart. How much longer, in inches, is bolt A than bolt B?



- A. $\frac{5}{8}$
 B. $1\frac{3}{8}$
 C. $1\frac{5}{8}$
 D. $1\frac{3}{4}$

28. At a fabric store, Melissa sold $8\frac{7}{8}$ yards of cloth to a customer. If the material was cut from a bolt of fabric containing $23\frac{1}{4}$ yards, how many yards are left on the bolt?

- A. $14\frac{3}{8}$
 B. $15\frac{3}{8}$
 C. $15\frac{3}{4}$
 D. $31\frac{7}{8}$

29. A batch of salad dressing requires $1\frac{2}{3}$ cups of olive oil, $\frac{1}{2}$ cup of vinegar, and $\frac{3}{4}$ cup of water. How many cups of salad dressing will this recipe produce?

- A. $1\frac{2}{3}$
 B. $2\frac{5}{6}$
 C. $2\frac{11}{12}$
 D. $3\frac{7}{12}$

Answers and explanations begin on page 663.

Multiplication and Division

It isn't necessary to find a common denominator to multiply and divide fractions. To multiply fractions, simply multiply the numerators and then the denominators. Reduce the answer, if necessary.

Example 5: What is the product of $\frac{7}{8}$ and $\frac{1}{2}$?

Multiply the numerators together and then the denominators. The answer is in lowest terms.

$$\frac{7}{8} \times \frac{1}{2} = \frac{7 \times 1}{8 \times 2} = \frac{7}{16}$$

Before multiplying a mixed number, change it to an improper fraction.

Example 6: What is $\frac{1}{3}$ of $3\frac{3}{4}$?

1. Change $3\frac{3}{4}$ to an improper fraction.

$$3\frac{3}{4} = \frac{15}{4}$$

2. Multiply the numerators and the denominators.

$$\frac{15}{4} \times \frac{1}{3} = \frac{15 \times 1}{4 \times 3} = \frac{15}{12}$$

3. Change to a mixed number and reduce to lowest terms.

$$\frac{15}{12} = 1\frac{3}{12} = 1\frac{1}{4}$$

You can use a shortcut called **canceling** to reduce the fractions as you work the problem. To cancel, divide both a numerator and a denominator by the same number. The numerator and the denominator can be in different fractions.

Example 7: Multiply $1\frac{1}{2}$ by $1\frac{1}{5}$.

1. Change to improper fractions.

$$1\frac{1}{2} = \frac{3}{2} \text{ and } 1\frac{1}{5} = \frac{6}{5}$$

2. Set up the multiplication problem. Both 6 (a numerator) and 2 (a denominator) are evenly divisible by 2. Divide them by 2. Then multiply using the new numerator and denominator. Finally, change the improper fraction to a mixed number.

$$\frac{3}{2} \times \frac{6}{5} = \frac{3}{\cancel{2}} \times \frac{\cancel{6}^3}{5} = \frac{9}{5} = 1\frac{4}{5}$$

The slash marks show that the numbers have been divided.

You will need two additional steps to divide fractions. Before dividing, **invert** the divisor (the fraction you are dividing by). To invert the fraction, switch the numerator and the denominator. Finally, change the division symbol to a multiplication symbol and multiply.

Example 8: Jim has an 8-pound bag of nuts. He wants to fill smaller $\frac{1}{2}$ -pound bags using the nuts. How many small bags can he make?

1. Divide 8 by $\frac{1}{2}$. Set up the division problem. Always write whole or mixed numbers as improper fractions.

$$8 \div \frac{1}{2} = \frac{8}{1} \div \frac{1}{2} =$$

2. Invert the fraction you are dividing by. Then change the operation sign to multiplication. Multiply, following the rules for multiplying fractions. Jim can make **16 small bags**.

$$\frac{8}{1} \times \frac{2}{1} = \frac{16}{1} = 16$$

NOTE: When you multiply by a fraction less than 1, the answer is smaller than the number you started with because you are finding a "part of." When you divide by a fraction less than 1, the answer is greater than the number.

Practice 4.2

A. Solve. Reduce answers to lowest terms. Simplify improper fractions as mixed numbers.

1. $\frac{2}{3} \times \frac{1}{4} =$
2. $1\frac{5}{6} \times \frac{1}{2} =$
3. $\frac{2}{3} \times 21 =$
4. $50 \times \frac{3}{8} =$
5. $3\frac{1}{2} \times \frac{1}{4} =$
6. $\frac{3}{4} \times \frac{7}{8} =$
7. $2\frac{1}{3} \times 3\frac{2}{5} =$
8. $15 \times 2\frac{3}{4} =$
9. $\frac{5}{8} \times 3\frac{1}{4} =$
10. $\frac{7}{8} \div \frac{1}{16} =$
11. $\frac{4}{5} \div \frac{4}{9} =$
12. $12 \div \frac{1}{4} =$
13. $6 \div 2\frac{1}{2} =$
14. $3\frac{3}{4} \div 1\frac{2}{3} =$
15. $9 \div \frac{1}{3} =$
16. $26\frac{2}{3} \div 3\frac{1}{3} =$
17. $40\frac{3}{8} \div 4\frac{1}{4} =$
18. $3\frac{7}{8} \div 5\frac{1}{6} =$

B. Choose the one best answer to each question.

19. A city is considering raising taxes to build a football stadium. A survey of registered voters yielded the following results:

Position	Fraction of Those Surveyed
Against Tax Hike	$\frac{7}{16}$
For Tax Hike	$\frac{3}{16}$
Undecided	$\frac{3}{8}$

If 400 people were surveyed, how many support the tax hike?

- A. 48
- B. 75
- C. 150
- D. 175

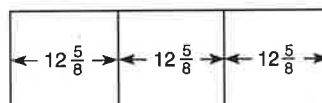
20. A tailor has 20 yards of shirt fabric. How many shirts can she complete if each shirt requires $2\frac{3}{4}$ yards of fabric?

- A. 6
- B. 7
- C. 8
- D. 10

21. An insurance agent estimates that it takes $\frac{2}{3}$ hour to process a customer's claim. If the agent spends 22 hours per week processing claims, about how many claims does he process in a week?

- A. $14\frac{2}{3}$
- B. 33
- C. 44
- D. 66

22. A fluorescent lighting panel is $12\frac{5}{8}$ inches wide. If three of the panels are installed as shown below, what will be the width in inches of the combined panels?



- A. $13\frac{7}{8}$
- B. $36\frac{5}{8}$
- C. $37\frac{7}{8}$
- D. $42\frac{7}{8}$

Answers and explanations begin on page 663.

Lesson 5: Fraction and Decimal Equivalencies

Fractions and decimals are two ways to show part of a whole. You can change fractions to decimals by dividing.

Example 1: Change $\frac{3}{8}$ to a decimal.

The fraction $\frac{3}{8}$ means $3 \div 8$. Use a calculator to divide.

$$3 \div 8 = 0.375$$

You can also change a decimal to a fraction.

Example 2: Change 0.35 to a fraction.

Write the decimal number over the place value name of the last decimal digit on the right. The last digit, 5, is in the hundredths column. Reduce to lowest terms.

$$\frac{35}{100} = \frac{35 \div 5}{100 \div 5} = \frac{7}{20}$$

You will find it useful to memorize the most common fraction and decimal equivalents. These equivalents will also help you solve percent problems.

Decimal	Fraction
0.1	$\frac{1}{10}$
0.125	$\frac{1}{8}$
0.2	$\frac{1}{5}$
0.25	$\frac{1}{4}$
0.3	$\frac{3}{10}$
0.33 $\overline{3}$	$\frac{1}{3}$

Decimal	Fraction
0.375	$\frac{3}{8}$
0.4	$\frac{2}{5}$
0.5	$\frac{1}{2}$
0.6	$\frac{3}{5}$
0.625	$\frac{5}{8}$
0.66 $\overline{6}$	$\frac{2}{3}$

Decimal	Fraction
0.7	$\frac{7}{10}$
0.75	$\frac{3}{4}$
0.8	$\frac{4}{5}$
0.875	$\frac{7}{8}$
0.9	$\frac{9}{10}$

GED® TEST TIP

Look at the answer choices before you begin working a multiple-choice problem. Knowing whether you need an answer in fraction or decimal form may affect how you approach the problem.

The decimal equivalents for $\frac{1}{3}$ and $\frac{2}{3}$ are marked with a bar. The bar shows that the decimal repeats indefinitely.

You can use fraction and decimal equivalents to save time when solving math problems.

Example 3: Each dose of cough medicine contains 0.25 ounce of medication. How many ounces of medication are in 48 doses?

To solve the problem, you need to multiply 48 by 0.25, a time-consuming calculation. However, since $0.25 = \frac{1}{4}$, you can find $\frac{1}{4}$ of 48 to solve the problem. The answer is **12 ounces**.

$$48 \times \frac{1}{4} = \frac{48}{4} = 12$$

Knowing fraction-decimal equivalents can also help you interpret remainders when using a calculator.

Example 4: Ray inspects machine assemblies. He must inspect 12 assemblies during his 40-hour workweek. On average, how many hours can he spend on each inspection?

Using a calculator, divide 40 by 12: 40 $\div$ 12 enter . The right side of the display reads 3.33333333.

Since you know that $0.33\overline{3} = \frac{1}{3}$, the answer is $3\frac{1}{3}$ hours.

Practice 5

Solve. When possible, use fraction and decimal equivalents to make the work easier. You MAY use a calculator on questions 5 and 6.

1. During a 25 percent-off sale, store clerks find the amount of the discounts by multiplying the regular price by 0.25. What is the discount on an item with a regular price of \$80?

A. \$32.00
B. \$20.00
C. \$16.40
D. \$2.00

2. At Linton Products, $\frac{3}{10}$ of the workers are in the company's ride-share program. If there are 480 workers, which of the following expressions could be used to find the number in the ride-share program?

A. 480×0.7
B. $480 \div 0.7$
C. 480×0.3
D. $480 \div 0.3$

3. Sharon is using a calculator to find out how many hours she has spent on a certain job. She divides, and her display reads:

4.66666666

Assuming her calculations are correct, how many hours did she spend on the job?

A. $4\frac{1}{6}$
B. $4\frac{2}{3}$
C. $4\frac{6}{7}$
D. 46

4. A gourmet candy company charges the following prices per pound.

Jelly Beans	\$9.60
Peanut Brittle	\$12.00
Almond Toffee	\$28.50

How much would a customer pay for 1.5 pounds of peanut brittle?

A. \$6.00
B. \$14.40
C. \$18.00
D. \$42.75

5. At 1 p.m., the amount of rain in a rain gauge is 1.125 inches. At 3 p.m., the gauge holds 1.875 inches. What fraction of an inch of rain fell between 1 p.m. and 3 p.m.?

A. $\frac{7}{8}$
B. $\frac{3}{4}$
C. $\frac{7}{10}$
D. $\frac{1}{8}$

6. A steel rod, 3 meters in length, is cut into 8 equal pieces. What is the length, in meters, of each piece?

A. 0.125
B. 0.333
C. 0.375
D. 2.333

Answers and explanations begin on page 663.